

Tirth Alkesh Patel

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SUMMARY

Curious data scientist who thrives at the intersection of ML, business strategy, and applied AI. Took a data-driven market thesis all the way to Shark Tank India auditions, defending revenue models and competitive analysis before executive. Delivers end-to-end ML systems in production across LLM engineering and cloud pipelines. MS Business Analytics at UCLA Anderson.

SKILLS

Analytics & Visualization: Tableau, Power BI, MS Excel (Pivot Tables, Power Query), Pandas, NumPy, Matplotlib, Google Sheets

Languages & Tools: Python, SQL, R, Git/GitHub, MySQL, Docker, MS PowerPoint, Claude Code, Oracle

Data Engineering: Apache Spark, PySpark, Snowflake, Apache Airflow, Google Cloud Storage, DBT, ETL Pipelines

ML & Modeling: scikit-learn, XGBoost, Random Forest, Regression, NLP (VADER/NLTK), PyTorch

Strategy & Research: Market Analysis, Competitive Intelligence, Market Sizing, Industry Research, Pricing Strategy

EDUCATION

UCLA ANDERSON SCHOOL OF MANAGEMENT

Los Angeles, CA

Master of Science in Business Analytics (MSBA)

Sep 2025 – Dec 2026

- Statistical Foundations of Analytics, Machine Learning, Data Engineering, Operations/Customer/Competitive Analytics

INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

Chennai, India

Bachelor of Technology, Chemical Engineering

Dec 2021 – May 2025

- Mathematical Foundations of Data Science, Advanced Operations Research, Operations Management, Economics

PROFESSIONAL EXPERIENCE

SEAT OF JOY – Stealth Startup

Chennai, India

Product Analyst

Aug 2023 – Feb 2024

- Analyzed 100+ Census of India 2011 tables using regression modeling to size a target market of 55M customers with 5% YoY growth, a 45% improvement over prior estimates, shaping the go-to-market strategy.
- Conducted SWOT analysis across 3 competitor categories, benchmarking pricing and positioning; called manufacturers to obtain cost data, calculated industry margins, and developed a competitive pricing strategy.
- Conducted 50+ customer interviews to surface behavioral patterns and pricing perceptions; translated qualitative findings into structured datasets that shaped product positioning and pricing decisions.

Product Analyst Intern

Jun 2023 – Jul 2023

- Built a strategic pitch deck integrating market sizing, competitive analysis, and revenue model; presented findings to executive judges at Shark Tank India auditions and advanced through 2 selection rounds.
- Designed an operations research model for state-wise product distribution across Indian states using demand and cost constraints; delivered a data-driven distribution strategy with actionable recommendations.

PROJECTS

Course RAG Pipeline

LangGraph, ChromaDB, FastAPI, LangChain, Docker, OpenAI API, Claude API

- Designed and deployed a production RAG system used by the UCLA MSBA cohort across 4 concurrent courses; architected 13 LangGraph nodes across 5 query-routing branches with output validation at each stage.
- Built a self-verifying deadline pipeline cross-referencing dual ChromaDB queries with LLM fallback; deployed on Oracle Cloud, resolving 9 production failures and adding a human-in-the-loop approval flow.

Weather & Dining Intelligence Pipeline

PySpark, Snowflake, Airflow, Tableau, VADER NLP

- Engineered a PySpark pipeline processing 10M+ Yelp reviews and NOAA weather data across 9 US cities via medallion architecture with VADER NLP; found freezing days drop volume 65% yet yield the highest sentiment index (0.71).
- Designed a Snowflake star schema enabling sub-second analytical queries; built an executive Tableau dashboard with KPI tiles and calculated fields to surface weather-driven consumer behavior insights.

Mapblazer Wait Time Prediction

Meta Prophet, XGBoost

- Architected an end-to-end ML pipeline benchmarking three forecasting architectures (Facebook Prophet, XGBoost Local, XGBoost Global) across 1.5M+ wait-time telemetry records from 6 Southern California theme parks.
- Cut prediction error 53.8% over the historical baseline (MAE: 7.08 to 3.27 min) with a fine-tuned Prophet model; achieved 90.38% error containment within ± 10 min and slashed severe misses by 62.9%.

Deep Learning Maltese Gear Cube Solver

PyTorch, CUDA, Deep Reinforcement Learning

- Built a deep reinforcement learning solver from the DeepCubeA paper, training a value network on 50M+ self-generated states in GPU-accelerated PyTorch/CUDA to learn a heuristic for a combinatorial puzzle.
- Reached a 100% solve rate at near-optimal move counts, then compressed the value network 4.5x (405M to 90M parameters) at the same solve rate, cutting training and inference compute by 10x.